

SECURE ENTERPRISE NETWORK & SOC LAB

EVE-NG Technical Project Documentation

Enterprise Networking • Network Security • SOC Monitoring

1. Project Overview

This project implements a secure enterprise campus network in EVE-NG. It combines VLAN segmentation, redundant Layer-3 switching, HSRP, OSPF, LACP EtherChannel, Rapid-PVST, DHCP, NTP, SSH, Port Security, DHCP Snooping, Dynamic ARP Inspection (DAI), Guest VLAN isolation, IDS, and centralized Syslog using a Windows 7 VM running Kiwi Syslog Server.

The lab demonstrates both network-engineering fundamentals and a practical security-monitoring workflow suitable for networking, SOC/Blue Team, and entry-level cloud-security interviews.

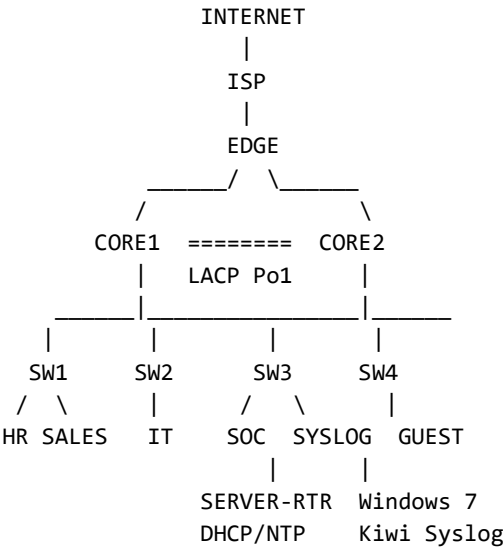
2. Project Objectives

- Build a segmented enterprise network using VLANs.
- Provide gateway redundancy using HSRP.
- Provide dynamic routing using OSPF.
- Provide resilient core connectivity using LACP EtherChannel.
- Implement Rapid-PVST and access-layer protections.
- Centralize DHCP and NTP services.
- Secure designated administrative access using SSH.
- Protect access ports using Port Security.
- Prevent rogue DHCP activity using DHCP Snooping.
- Validate ARP traffic using Dynamic ARP Inspection.
- Isolate Guest users from internal enterprise VLANs.
- Detect configured security events using IDS.
- Centralize device logs in Kiwi Syslog for SOC-style monitoring.

3. Lab Environment

Component	Purpose
EVE-NG	Network emulation platform
Cisco routers	ISP, EDGE and SERVER-RTR
Cisco Layer-3 switches	CORE1 and CORE2; routing, SVIs and HSRP
Cisco Layer-2 switches	SW1–SW4; access connectivity and security
Windows 7 VM	Centralized Syslog host running Kiwi Syslog Server 9.6.5 Legacy
Kali Linux	Security testing / validation host
Kiwi Syslog Server	Centralized Cisco Syslog collection and filtering

4. Logical Topology



5. Device and Interface Mapping

Device	Interface	Connected To	Role
ISP	Gi0/0	EDGE Gi0/0	Internet-side router
EDGE	Gi0/0	ISP Gi0/0	WAN edge
EDGE	Gi0/1	CORE1 Gi0/0	OSPF/core uplink
EDGE	Gi0/2	CORE2 Gi0/0	OSPF/core uplink
CORE1	Gi0/1	CORE2 Gi0/1	LACP member
CORE1	Gi0/2	CORE2 Gi0/2	LACP member
CORE1	Gi0/3	SW1 Gi0/0	Access uplink
CORE2	Gi0/3	SW1 Gi0/1	Access uplink
CORE1	Gi1/0	SW2 Gi0/0	Access uplink
CORE2	Gi1/0	SW2 Gi0/1	Access uplink
CORE1	Gi1/1	SW3 Gi0/0	Access uplink
CORE2	Gi1/1	SW3 Gi0/1	Access uplink
CORE1	Gi1/2	SW4 Gi0/0	Access uplink
CORE2	Gi1/2	SW4 Gi0/1	Access uplink
SW3	Gi0/2	SERVER-RTR Gi0/0	DHCP/NTP path
SW3	Gi1/0	Windows 7	Centralized Syslog

6. VLAN and IP Addressing Plan

VLAN	Name	Subnet	HSRP VIP	CORE1	CORE2
10	HR	10.10.10.0/24	10.10.10.1	10.10.10.2	10.10.10.3
20	SALES	10.10.20.0/24	10.10.20.1	10.10.20.2	10.10.20.3
30	IT	10.10.30.0/24	10.10.30.1	10.10.30.2	10.10.30.3
40	SERVER	10.10.40.0/24	10.10.40.1	10.10.40.2	10.10.40.3
50	GUEST	10.10.50.0/24	10.10.50.1	10.10.50.2	10.10.50.3
60	SOC	10.10.60.0/24	10.10.60.1	10.10.60.2	10.10.60.3
99	MANAGEMENT	10.10.99.0/24	10.10.99.1	10.10.99.2	10.10.99.3

Infrastructure links:

Link	Addressing
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ISP–EDGE	203.0.113.0/30 — ISP .1, EDGE .2
EDGE–CORE1	10.255.0.0/30 — EDGE .1, CORE1 .2
EDGE–CORE2	10.255.0.4/30 — EDGE .5, CORE2 .6
CORE1–CORE2 Po1	10.255.0.8/30 — CORE1 .9, CORE2 .10
SERVER-RTR Gi0/0	10.10.40.10/24
Windows 7 Syslog	10.10.40.20/24

7. Routing and High Availability

OSPF Area 0 runs between EDGE, CORE1 and CORE2. Router IDs are EDGE 3.3.3.3, CORE1 1.1.1.1 and CORE2 2.2.2.2. EDGE has a default route toward the simulated ISP and advertises the default route into OSPF.

HSRP provides a virtual default gateway for each VLAN. CORE1 is the primary role and CORE2 is the secondary path, providing gateway availability during a core failure.

8. Switching and Redundancy

- Rapid-PVST is used for spanning-tree control.
- CORE1 is STP root primary and CORE2 is secondary.
- CORE1–CORE2 uses LACP EtherChannel (Port-channel 1).
- Access uplinks are redundant through both cores.
- PortFast and BPDU Guard are applied to appropriate end-device ports.
- Port Security is applied to PC-facing access ports with sticky MAC learning and maximum one MAC address.

9. DHCP and NTP

SERVER-RTR provides centralized DHCP service. Core SVIs use the relay address 10.10.40.10. SERVER-RTR also provides NTP using ntp master 3; EDGE, CORE1 and CORE2 synchronize to 10.10.40.10. Accurate time supports Syslog correlation.

10. Security Controls

Control	Location	Purpose
Port Security	PC-facing access ports	Limits unauthorized MAC addresses and restricts violations.
DHCP Snooping	Access switches	Builds bindings and helps block rogue DHCP servers.
Dynamic ARP Inspection	Access VLANs	Validates ARP against DHCP snooping information.
BPDU Guard + PortFast	End-device ports	Protects access ports from unexpected STP BPDUs.
Guest ACL	VLAN 50 SVI	Blocks Guest-to-internal VLAN traffic while permitting allowed external traffic.
IDS	Configured IDS device	Detects configured suspicious/security traffic and generates events.
Centralized Syslog	Cisco devices → 10.10.40.20	Centralizes operational and security logs.

SSH	ISP and EDGE	Provides secure remote administration on designated devices.
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11. Guest VLAN Security

VLAN 50 uses the GUEST extended ACL on the Layer-3 core SVI. Traffic from 10.10.50.0/24 to HR, SALES, IT, SERVER, SOC and MANAGEMENT subnets is blocked. DHCP is explicitly permitted and the final rule permits allowed external traffic. The ACL is configured on both CORE1 and CORE2 so the policy remains available during HSRP failover.

12. DHCP Snooping and DAI

Infrastructure-facing and DHCP-server-facing interfaces are trusted; end-device ports are untrusted. PC-facing ports use a DHCP packet rate limit of 10 packets per second. DHCP Option 82 is disabled in this lab because it caused compatibility issues in the EVE-NG environment.

DAI is enabled on the relevant access VLANs. Core uplinks and the DHCP-server path are trusted, while end-device ports remain untrusted. DAI uses DHCP snooping bindings to validate ARP information.

13. Centralized Syslog and SOC Monitoring

Windows 7 at 10.10.40.20 runs Kiwi Syslog Server 9.6.5 Legacy and receives centralized Cisco Syslog messages over UDP 514. Local buffered logging is also enabled on Cisco devices.

```
logging host 10.10.40.20
logging trap informational
logging buffered 16384 informational
service timestamps log datetime msec
service timestamps debug datetime msec
```

Kiwi filters were configured for interface UP/DOWN events and security login events (SEC_LOGIN), providing a basic SOC-style filtering workflow.

14. IDS Workflow

The IDS configuration has been completed and tested successfully in the lab. The operational workflow is:

- A test host generates traffic against a monitored path.
- The IDS evaluates the traffic against configured detection/signature rules.
- A matching event generates an IDS alert.
- Where configured, the network device generates associated Syslog information.
- Kiwi Syslog receives and displays the centralized log.
- A SOC analyst can correlate source, destination, timestamp and other available logs.

15. Validation and Test Results

Test	Expected Result	Result
Inter-VLAN connectivity	Authorized VLANs communicate through cores	Working

HSRP	Virtual gateway remains available during core failover	Working
OSPF	Routes learned dynamically	Working
EtherChannel	Core links operate as logical Port-channel	Working
DHCP	Clients receive addresses from SERVER-RTR	Working
NTP	Devices synchronize to SERVER-RTR	Working
Port Security	Access ports enforce MAC policy	Working
DHCP Snooping	Bindings are learned and rogue DHCP is mitigated	Working
DAI	ARP is validated using snooping information	Working
Guest isolation	Guest cannot access protected internal VLANs	Working
Syslog	Cisco logs arrive at Kiwi Syslog	Working
IDS	Configured IDS detection operates and produces events	Working

16. Useful Verification Commands

```

show ip interface brief
show ip ospf neighbor
show ip route ospf
show standby brief
show etherchannel summary
show spanning-tree
show ip dhcp binding
show ip dhcp pool
show ntp status
show logging
show port-security
show ip dhcp snooping
show ip dhcp snooping binding
show ip arp inspection
show ip arp inspection interfaces
show ip arp inspection statistics
show ip ips configuration
show ip ips interfaces
show ip ips statistics

```

17. SOC Event Demonstration

```

Test traffic
↓
Network security control / IDS
↓
Detection event
↓
Cisco logging
↓
UDP 514
↓
Kiwi Syslog Server

```

↓

SOC analyst reviews source, destination, timestamp and event

18. Interview Explanation

"I built a secure enterprise network in EVE-NG with a redundant Layer-3 core. The network is segmented into HR, Sales, IT, Server, Guest, SOC and Management VLANs. CORE1 and CORE2 provide gateway redundancy using HSRP, OSPF provides dynamic routing, and LACP EtherChannel provides resilient core connectivity. Rapid-PVST is used for Layer-2 loop prevention.

For security, I implemented Port Security, DHCP Snooping and Dynamic ARP Inspection on the access layer, along with BPDU Guard and PortFast on end-device ports. The Guest VLAN is isolated with an extended ACL so Guest users cannot reach internal enterprise networks.

For monitoring, SERVER-RTR provides DHCP and NTP, while a Windows 7 VM running Kiwi Syslog Server receives centralized Syslog messages from the Cisco devices. I also configured and tested IDS so security events can be detected and investigated. The project demonstrates prevention, detection, logging and basic SOC investigation workflows."

19. Skills Demonstrated

- Cisco IOS configuration and troubleshooting
- VLANs, trunking and inter-VLAN routing
- OSPF and default-route propagation
- HSRP and network redundancy
- LACP EtherChannel
- Rapid-PVST
- DHCP and DHCP relay
- NTP and time synchronization
- SSH administration
- ACLs
- Port Security
- DHCP Snooping
- Dynamic ARP Inspection
- IDS concepts and validation
- Centralized Syslog and Kiwi Syslog
- SOC-style event monitoring
- EVE-NG network emulation

20. Conclusion

The completed lab provides an end-to-end secure enterprise network with redundancy, segmentation, access-layer protections, centralized services, intrusion detection and centralized logging. It can be reused for troubleshooting, security validation and SOC investigation practice.

End of Project Documentation